

HemanthKumar K Ramakrishnan

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SUMMARY

Lead Software Engineer with 8 years of experience designing scalable distributed systems, event-driven microservices, and full-stack applications using Java, Python, TypeScript/React, Spring WebFlux, AWS, Kafka, Spark, and MongoDB. Experienced in building high-throughput data pipelines and cloud-native applications on Kubernetes/EKS. Hands-on with Generative and Agentic AI using LangGraph, MCP, RAG, multi-agent orchestration, tool calling, and LLM-based automation, with a focus on integrating AI agents into enterprise engineering workflows.

SKILLS

Java, Python, PySpark, Spark, R | TypeScript, React | SQL, No-SQL | Spring Boot, Spring WebFlux | AWS | Kafka | Kubernetes, Docker | Redis, MongoDB | Agentic AI, LangGraph, MCP, RAG, Tool Calling, Multi-Agent Systems, LangSmith | Informatica, Control-M | Splunk, Kibana

EXPERIENCE

Lead Software Engineer

Freddie Mac (Client) - Thinklusive Inc (FTE)

May 2020 - Present, McLean, Virginia

- Led delivery of a web application built on Java 17, Spring WebFlux and TypeScript with React for processing 1M+ appraisals/month on AWS EKS.
- Built Java backend application to process XML files in S3 via SQS poll and convert to JSON, enrich via external APIs and store in MongoDB.
- Designed File uploader component using React, TypeScript and styled with Tailwind CSS, cutting upload errors by 75% and latency by 30%.
- Delivered multiple GraphQL & gRPC APIs over MongoDB, secured with Apigee & OAuth to deliver sub-second responses adopted by 27+ teams.
- Built Agentic AI workflows using LangGraph, MCP, and tool calling to automate Jira story creation, task execution, code review, testing, and PR.
- Developed RAG and incident-triage agents with LangSmith tracing to answer project queries, summarize Jenkins failures, and create L3 incidents.
- Developed scalable PySpark jobs on AWS EMR to enrich 1M+ records per batch with scrubbed addresses and external metadata sources.
- Automated daily NAS-to-S3 file transfers using Control-M file watchers and Informatica CDI, completely removing manual processing steps.
- Deployed interactive Kibana dashboards with real-time pod health and error log alerts to notify L3 and business teams during incidents.

Software Engineer

Labcorp (Client) - Thinklusive Inc (FTE)

June 2019 - May 2020, Durham, North Carolina

- Developed a robust Java Spring Boot application to retrieve patient and service center data from a DB2 database for healthcare reporting.
- Implemented caching using Spring Cache and Redis to reduce database calls and improve response time for frequently accessed endpoints.
- Built scalable R Shiny web app on Cloudera for self-service dashboards, increasing stakeholder satisfaction by 40% with real-time insights.
- Migrated legacy data from Oracle database to Hive using Sqoop and HiveQL, significantly reducing batch runtimes from 45 to 12 minutes.
- Automated datameer workbook generation for genetics analytics, cutting report delivery from 3 days to under 2 hours for stakeholders.

Software Engineer

VentureSpace

August 2018 - May 2019, New York, New York

- Built robust Python web scrapers aggregating listings from 10+ sites daily, tripling data coverage and reducing manual research time by 80%.
- Deployed scalable FastAPI microservice with Pydantic models and 95% test coverage, delivering live valuations under 50 ms for broker tools.
- Optimized complex PostgreSQL queries and materialized views, cutting dashboard load times from 12s to 1.5s for real-time trend analysis.
- Created 15+ interactive dashboards in Superset and Tableau, surfacing key business metrics that drove a 20% boost in listing conversions.

EDUCATION

Master of Science : Information Systems

New Jersey Institute of Technology • New Jersey, USA • 2019 • 3.65 GPA

Bachelors of Engineering : Electronics and Communication Engineering

Anna University • Chennai, India • 2017 • 3.2 GPA

CERTIFICATIONS

AWS Certified Cloud Developer

AWS

Claude Certified Architect (CCAR-F)

Anthropic • 2026
